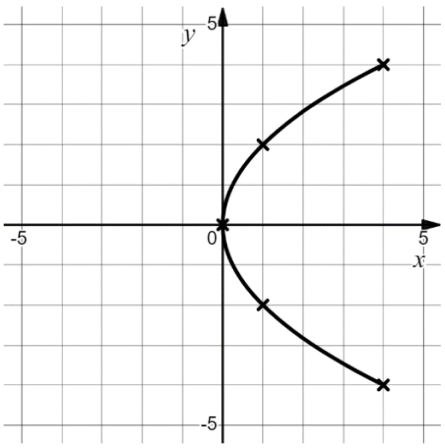
2.41 Parametric Equations: Introduction

Worked Solutions 1–21

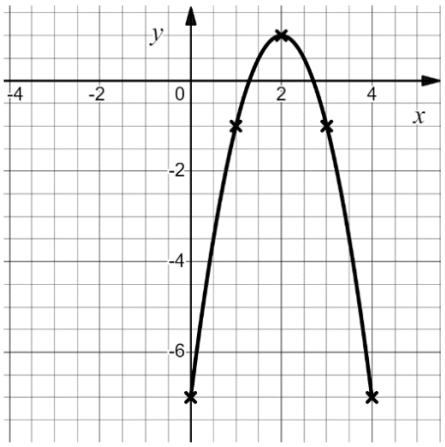
1 a)

t	-2	-1	0	1	2
x	4	1	0	1	4
y	-4	-2	0	2	4



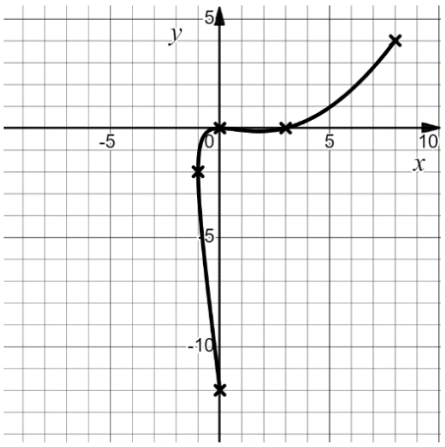
1 b)

t	-2	-1	0	1	2
x	0	1	2	3	4
y	-7	-1	1	-1	-7



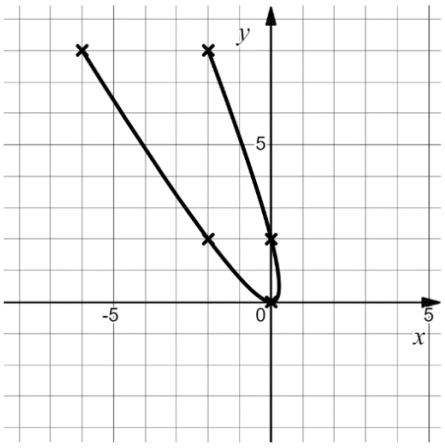
1 c)

t	-2	-1	0	1	2
x	0	-1	0	3	8
y	-12	-2	0	0	4



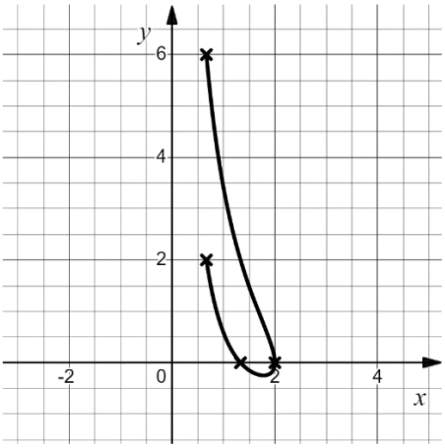
1 d)

t	-2	-1	0	1	2
x	-6	-2	0	0	-2
y	8	2	0	2	8



1 e)

t	-2	-1	0	1	2
x	$\frac{2}{3}$	$\frac{4}{3}$	2	$\frac{4}{3}$	$\frac{2}{3}$
y	2	0	0	0	6



2 a)

$y = 2t^2$.

2 b)

$y = t^2 + 4$.

2 c)

$y = e^t + 3$.

2 d)

$y = 8 + 2t$.

2 e)

$y = t^2$.

2 f)

$y = -\frac{10}{t^3}$.

2 g)

$x = \cos t$, $y = \sin t$ or $x = \sin t$, $y = \cos t$.

3 a)

$t = \frac{x}{2}$ so $y = (\frac{x}{2})^2 \Rightarrow y = \frac{1}{4}x^2$.

3 b)

$t = x + 1$ so $y = 2(x + 1) + 3 \Rightarrow y = 2x + 5$.

3 c)

$t = x + 3$ so $y = e^{x+3} - x - 3$.

3 d)

$t = -\frac{x}{2}$ so $y = \frac{1}{4}x^2 \ln(-\frac{x}{2})$.

4 a)

$y = 1 + t$ so $x = (1 + y)^2$.

4 b)

$t = 2y$ so $x = (2y)^3 - 2y \Rightarrow x = 8y^3 - 2y$.

4 c)

$t = \frac{y}{3}$ so $x = (\frac{y}{3})^5 \Rightarrow x = \frac{1}{243}y^5$.

4 d)

$t = y - 5$ so $x = e^{y-6}$.

5 a)

$t = -1$ gives $(0, -4)$. $t = 3$ gives $(4, 0)$.

5 b)

$t = \frac{\pi}{2}$, $t = \frac{3\pi}{2}$ gives $(0, 2)$ and $(0, -2)$. $t = 0$, $t = \pi$ gives $(1, 0)$ and $(-1, 0)$.

5 c)

$t = \frac{\pi}{2}$, $t = \frac{3\pi}{2}$ gives $(0, -1)$ and $(0, 1)$. $t = 0$, $t = \frac{\pi}{3}$, $t = \frac{2\pi}{3}$, $t = \pi$ gives $(2, 0)$, $(1, 0)$, $(-1, 0)$ and $(-2, 0)$.

5 d)

$t = 0$ gives $\ln 1 = 0$ so $(5, 0)$. $t = -5$ gives $\ln -4$ which doesn't exist so it doesn't cross the x-axis.

5 e)

$t = 1$, $t = -2$, $t = 4$ gives $(0, e^{-2})$, $(0, e^4)$ and $(0, e^{-8})$. It doesn't cross the x-axis.

6)

$y^2 = (t^3)^2 = t^6 = (t^2)^3 = x^3$ so $k = 3$.

7 a)

$$\begin{aligned} x^2 &= (t + \tfrac{1}{t})(t + \tfrac{1}{t}) = t^2 + 2 + \tfrac{1}{t^2} \\ y^2 &= (t - \tfrac{1}{t})(t - \tfrac{1}{t}) = t^2 - 2 + \tfrac{1}{t^2} \\ \text{So } x^2 - y^2 &= t^2 + 2 + \tfrac{1}{t^2} - (t^2 - 2 + \tfrac{1}{t^2}) = 4 \\ x^2 - y^2 &= 4 \end{aligned}$$

7 b)

$$\begin{aligned} x^2 &= (t-\tfrac{2}{t})(t-\tfrac{2}{t}) = t^2 - 4 + \tfrac{4}{t^2} \\ y^2 &= (t+\tfrac{2}{t})(t+\tfrac{2}{t}) = t^2 + 4 + \tfrac{4}{t^2} \\ \text{So } x^2 - y^2 &= t^2 - 4 + \tfrac{4}{t^2} - (t^2 + 4 + \tfrac{4}{t^2}) = -8 \\ x^2 - y^2 &= -8 \end{aligned}$$

7 c)

$$\begin{aligned} x^2 &= (7t+\tfrac{4}{t})(7t+\tfrac{4}{t}) = 49t^2 + 56 + \tfrac{16}{t^2} \\ y^2 &= (7t-\tfrac{4}{t})(7t-\tfrac{4}{t}) = 49t^2 - 56 + \tfrac{16}{t^2} \\ \text{So } x^2 - y^2 &= 49t^2 + 56 + \tfrac{16}{t^2} - (49t^2 - 56 + \tfrac{16}{t^2}) = 112 \\ x^2 - y^2 &= 112 \end{aligned}$$

8)

$x + 1 = \frac{1}{t}$ so $t = \frac{1}{x+1}$.

$$y = \frac{2 + 3(\frac{1}{x+1})}{1 + 3(\frac{1}{x+1})} = \frac{2(x+1)+3}{(x+1)+3} = \frac{2x+5}{x+4} \text{ so } a = 2, b = 5$$

9 a)

$(2x)^2 + (3y)^2 = 1$. $2x = \cos\theta$, $3y = \sin\theta$ so $x = \frac{1}{2}\cos\theta$, $y = \frac{1}{3}\sin\theta$. OR $2x = \sin\theta$, $3y = \cos\theta$ so $x = \frac{1}{2}\sin\theta$, $y = \frac{1}{3}\cos\theta$.

9 b)

$(\sqrt{3}x)^2 + (\sqrt{2}y)^2 = 1$. $\sqrt{3}x = \cos\theta$, $\sqrt{2}y = \sin\theta$ so $x = \frac{1}{\sqrt{3}}\cos\theta$, $y = \frac{1}{\sqrt{2}}\sin\theta$. OR $\sqrt{3}x = \sin\theta$, $\sqrt{2}y = \cos\theta$ so $x = \frac{1}{\sqrt{3}}\sin\theta$, $y = \frac{1}{\sqrt{2}}\cos\theta$.

9 c)

$(\sqrt{5}x)^2 + (\frac{1}{\sqrt{2}}y)^2 = 1$. $\sqrt{5}x = \cos\theta$, $\frac{1}{\sqrt{2}}y = \sin\theta$ so $x = \frac{1}{\sqrt{5}}\cos\theta$, $y = \sqrt{2}\sin\theta$. OR $\sqrt{5}x = \sin\theta$, $\frac{1}{\sqrt{2}}y = \cos\theta$ so $x = \frac{1}{\sqrt{5}}\sin\theta$, $y = \sqrt{2}\cos\theta$.

9 d)

$2x^2 + \frac{1}{3}y^2 = 1$. $(\sqrt{2}x)^2 + (\frac{1}{\sqrt{3}}y)^2 = 1$. $\sqrt{2}x = \cos\theta$, $\frac{1}{\sqrt{3}}y = \sin\theta$ so $x = \frac{1}{\sqrt{2}}\cos\theta$, $y = \sqrt{3}\sin\theta$. OR $\sqrt{2}x = \sin\theta$, $\frac{1}{\sqrt{3}}y = \cos\theta$ so $x = \frac{1}{\sqrt{2}}\sin\theta$, $y = \sqrt{3}\cos\theta$.

10 a)

$\frac{x}{3} = \cos\theta$ and $\frac{y}{3} = \sin\theta$. Since $\sin^2\theta + \cos^2\theta = 1$, $(\frac{y}{3})^2 + (\frac{x}{3})^2 = 1$. $x^2 + y^2 = 9$.

10 b)

$\frac{x}{5} = \sin\theta$ and $\frac{y}{-5} = \cos\theta$. Since $\sin^2\theta + \cos^2\theta = 1$, $(\frac{x}{5})^2 + (\frac{y}{-5})^2 = 1$. $x^2 + y^2 = 25$.

10 c)

$x - 1 = \cos\theta$ and $y + 3 = \sin\theta$. Since $\sin^2\theta + \cos^2\theta = 1$, $(y+3)^2 + (x-1)^2 = 1$.

10 d)

$x + 4 = \sin\theta$ and $y - 8 = \cos\theta$. Since $\sin^2\theta + \cos^2\theta = 1$, $(x+4)^2 + (y-8)^2 = 1$.

11 a)

$\frac{x}{10} = \cos t$ and $\frac{y-15}{5} = \sin t$. Since $\sin^2 t + \cos^2 t = 1$, $(\frac{y-15}{5})^2 + (\frac{x}{10})^2 = 1$. $4(y-15)^2 + x^2 = 100$.

11 b)

Maximise $y \Rightarrow y = 15 + 5 = 20$ m.

11 c)

$\sin t = 1 \Rightarrow t = 90°$ or $t = \frac{\pi}{2}$.

11 d)

2π.

12 a)

$\frac{x-2k}{2k} = \cos\theta$ and $\frac{y-8}{-3} = \sin\theta$. Since $\sin^2\theta + \cos^2\theta = 1$,

$$(\tfrac{y-8}{-3})^2 + (\tfrac{x-2k}{2k})^2 = 1$$
$$(y-8)^2 + 9(\tfrac{x-2k}{2k})^2 = 9$$
$$9(x-2k)^2 + 4k^2(y-8)^2 = 36k^2$$

12 b)

$4k^2 = 9 \Rightarrow k^2 = \frac{9}{4} \Rightarrow k = \frac{3}{2}$.

12 c)

$(2k, 8) = (3, 8)$.

12 d)

$r^2 = \frac{36}{9}k^2 \Rightarrow r^2 = \frac{36}{9} \times \frac{9}{4} = 9$, therefore $r = 3$.

13)

$\frac{x}{2} = \tan\theta$ and $y - 2 = \sec\theta$. Since $\tan^2\theta + 1 = \sec^2\theta$, $(\frac{x}{2})^2 + 1 = (y-2)^2$.

14 a)

Q is when $x = \frac{3}{2}$ so $\frac{3}{2} = \sqrt{3}\sin\theta$ giving $\theta = \frac{\pi}{3}$.

$$y = \sin(2 \times \tfrac{\pi}{3}) = \frac{\sqrt{3}}{2} \text{ so } Q(\tfrac{3}{2}, \tfrac{\sqrt{3}}{2})$$

P is when $y = \frac{\sqrt{3}}{2}$ so $\sin 2\theta = \frac{\sqrt{3}}{2}$ giving $\theta = \frac{\pi}{6}$.

$$x = \sqrt{3}\sin\tfrac{\pi}{6} = \frac{\sqrt{3}}{2} \text{ so } P(\tfrac{\sqrt{3}}{2}, \tfrac{\sqrt{3}}{2})$$

14 b)

Area is $\frac{\sqrt{3}}{2}(\frac{3}{2} - \frac{\sqrt{3}}{2}) = -\frac{3}{4} + \frac{3}{4}\sqrt{3} = \frac{3}{4}(\sqrt{3} - 1)$ units^2.

15 a)

$x^2 = \cos^2\theta = 1 - \sin^2\theta$. $x^2 = 1 - y$ or $y = 1 - x^2$.

15 b)

If $x = \sin\theta$ then using pythagoras' theorem: $\tan\theta = \frac{x}{\sqrt{1-x^2}}$ so $y = \frac{x}{\sqrt{1-x^2}}$.

16 a)

$$\begin{aligned}
(x-4)^2 + y^2 &= (\tfrac{t^2+7}{t^2+1} - 4)^2 + (\tfrac{6t}{t^2+1})^2 \\
&= (\tfrac{t^2+7}{t^2+1} - \tfrac{4(t^2+1)}{t^2+1})^2 + (\tfrac{6t}{t^2+1})^2 \\
&= (\tfrac{-3t^2+3}{t^2+1})^2 + (\tfrac{6t}{t^2+1})^2 \\
&= \frac{9t^4 - 18t^2 + 9 + 36t^2}{(t^2+1)^2} = \frac{9t^4 + 18t^2 + 9}{(t^2+1)^2} \\
&= \frac{9(t^4 + 2t^2 + 1)}{(t^2+1)^2} = \frac{9(t^2+1)}{t^2+1} = 9
\end{aligned}$$

So $(x-4)^2 + y^2 = 9$.

16 b)

Radius is 3.

17 a)

$$\begin{aligned}
(10\cos\theta)^2 + (4\sin\theta)^2 &= 100 \\
100\cos^2\theta + 16\sin^2\theta &= 100 \\
25(1 - \sin^2\theta) + 4\sin^2\theta &= 25 \\
25 - 25\sin^2\theta + 4\sin^2\theta &= 25 \\
\sin^2\theta &= 0 \text{ so } \theta = 0, \pi
\end{aligned}$$

Which gives $(10, 0)$ and $(-10, 0)$.

17 b)

$$\begin{aligned}
(2\cos\theta)^2 + (10\sin\theta)^2 &= 100 \\
4\cos^2\theta + 100\sin^2\theta &= 100
\end{aligned}$$

$$(1-\sin^2\theta)+25\sin^2\theta = 25$$
$$24\sin^2\theta = 24$$
$$\sin^2\theta = 1 \text{ so } \theta = \tfrac{\pi}{2}, \tfrac{3\pi}{2}$$

Which gives $(0, 10)$ and $(0, -10)$.

17 c)

$$(6\sqrt{2}\cos\theta)^2 + (2\sqrt{46}\sin\theta)^2 = 100$$
$$72\cos^2\theta + 184\sin^2\theta = 100$$
$$18(1-\sin^2\theta) + 46\sin^2\theta = 25$$
$$28\sin^2\theta = 7$$
$$\sin^2\theta = \tfrac{1}{4} \text{ so } \sin^2\theta = \pm\tfrac{1}{2}$$

So $\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ which gives $(3\sqrt{6}, 3\sqrt{5})$. By symmetry: $(-3\sqrt{6}, 3\sqrt{5})$, $(3\sqrt{6}, -3\sqrt{5})$ and $(-3\sqrt{6}, -3\sqrt{5})$. So the points of intersection are $(3\sqrt{6}, 3\sqrt{5})$, $(-3\sqrt{6}, 3\sqrt{5})$, $(3\sqrt{6}, -3\sqrt{5})$ and $(-3\sqrt{6}, -3\sqrt{5})$.

18 a)

$y = -1+2\cos 2\theta = -1+2(2\cos^2\theta-1)$. $y = 4\cos^2\theta-3$ but $\cos\theta = \frac{1}{2}(x-5)$. So $y = 4(\frac{1}{2}(x-5))^2-3 = (x-5)^2-3$.

18 b)

$0 \le \theta \le \pi$ so $5-2 \le x \le 5+2$. Domain is $\{x : x \in \mathbb{R},\ 3 \le x \le 7\}$. $0 \le \theta \le \pi$ so $-1-2 \le x \le -1+2$. Range is $\{y : y \in \mathbb{R},\ -3 \le y \le 1\}$.

18 c)

$$y = (x-5)^2 - 3 \text{ and } y = k - x$$
$$k - x = (x-5)^2 - 3$$
$$k - x = x^2 - 10x + 25 - 3$$
$$x^2 - 9x + (22-k) = 0$$

Two distinct solutions: $b^2 - 4ac > 0$. $81 - 4(22-k) > 0$. $4k > 7$ so $k > \frac{7}{4}$. But the last point at which there are two solutions is at $(3, 1)$. $1 = k - 3$ giving $k = 4$. So $\frac{7}{4} < k < 4$.

19 a)

Domain: $x \ge 1$ Range: $y \ge 0$.

19 b)

Domain: $x \ge 0$ Range: $y \le 4$.

19 c)

Domain: $-1 \le x \le 1$ Range: $-1 \le y \le 1$.

19 d)

Domain: $-1 \leq x \leq 1$ Range: $-1 \leq y \leq 1$.

19 e)

Domain: $x > 0$ Range: $y > 0$.

19 f)

Domain: $x \in \mathbb{R}$ Range: $y \geq 0$.

20)

$t \geq 0$ due to the $\sqrt{t}$. Domain: $-1 \leq x \leq 1$ Range: $y > 0$.

21 a)

Domain: $-1 \leq x \leq 1$ Range: $-4 \leq y \leq 4$.

21 b)

When $x = 0$, $\theta \in \{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\}$. So $(0, 2\sqrt{3})$, $(0, 2)$, $(0, -2\sqrt{3})$ and $(0, -2)$. When $y = 0$, $\theta = \frac{2\pi}{3}$ so $(-\frac{\sqrt{3}}{2}, 0)$.

21 c)

